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Portfolio: walt1210.github.io



CAREER PROFILE

Fourth year Computer Science student majoring in Data Science. Thesis work included building the retrieval-augmented guidance backend on Google Cloud Run and contributing to the model pipeline and Android application of an offline maize disease diagnosis system.

SKILLS

- **Programming Languages:** Python, R, Java, C#, PHP, JavaScript, Kotlin, Dart, SQL, HTML, CSS, Assembly
- **Machine Learning and Data Science:** TensorFlow, PyTorch, scikit-learn, Pandas, NumPy, NLTK, HDBSCAN, Matplotlib, Seaborn, Power BI
- **Agentic AI and LLM Frameworks:** LangChain, LangGraph (multi-agent workflows, human approval), CrewAI, Model Context Protocol (MCP), Google Gemini API, Pydantic (structured LLM output), RAG pipeline design, prompt engineering, Ollama, LLM evaluation
- **Web, Mobile, and Backend Development:** Flask, Android (Kotlin), Flutter, Next.js 15, Supabase
- **Databases:** MySQL, SQLite, Oracle, ChromaDB (vector database)
- **Cloud and DevOps:** Docker, Google Cloud Run, Google Cloud Platform, Google Colab, Linux, Git, Jupyter Notebook, Visual Studio Code

PROJECT EXPERIENCE

MAIze: AI-Powered Maize Disease Diagnosis System (Guidance Module) *Thesis Project*

Developed a retrieval-augmented generation system that grounds AI-generated crop disease management advice in a curated agricultural knowledge base. Deployed as a cloud backend supporting a mobile diagnostic application for Filipino farmers.

Role: NLP and Backend Developer. Designed and deployed the retrieval-augmented generation pipeline, conducted prompt engineering, developed the evaluation methodology, and managed cloud infrastructure.

Technologies: Python, Flask, LangChain, Google Gemini API, ChromaDB, RAGAS, Pydantic, Docker, Google Cloud Run | **Repository:** **github.com/walt1210/maize_nlp**

YellowMAIze: Mobile-Deployed Maize Disease Diagnosis System *Thesis Project*

Multi-task deep learning model for maize leaf disease classification, segmentation, and severity estimation, run on-device in TensorFlow Lite.

Role: Helped develop and improve the model pipeline, served as one of two severity raters, and worked on parts of the Android app.

Technologies: PyTorch, TensorFlow Lite, ONNX, scikit-learn, Grad-CAM family, Cohen's Kappa, Spearman Correlation | **Repository:** **github.com/LunocoreWired/yellowmaize**

BayanBench: Auditing AI Knowledge of Philippine Towns *Independent Project*

Built a benchmark of 7,441 questions from the official Philippine Standard Geographic Code (1,642 cities and municipalities, 42,010 barangays) to test whether AI models answer less accurately about small and remote towns, and tested a retrieval fix using official records.

Role: Sole developer. Built the question set and automatic scorer (49 of 49 verdicts confirmed by hand), ran five models, and analyzed errors with logistic regression clustered by town.

Technologies: Python, pandas, NumPy, Ollama, Gemini API, Matplotlib | **Repository:** **github.com/walt1210/bayanbench**

EventOps AI: Governed Multi-Agent Event Planning *Academic Project, Group*

Four LangGraph agents turn a one-sentence request into a school event plan, with Python costing and required human approval.

Role: Built the project notebook for the group: agents, orchestration, seven governance rules, and tests (7 of 7 passed).

Technologies: Python, LangGraph, Google Gemini API, Pydantic, pandas, ipywidgets | **Repository:** **github.com/walt1210/eventops-ai**

EDUCATION

Angeles University Foundation, Bachelor of Science in Computer Science, Major in Data Science (Expected Graduation: 2027)

CERTIFICATIONS

- Reinventing Learning through Artificial Intelligence, Holy Angel University
- Introduction to Cybersecurity, Cisco Networking Academy
- Introduction to Data Science, Cisco Networking Academy
- Microsoft Azure Fundamentals (AZ-900)
- Discovering Entrepreneurship, Cisco Networking Academy
- CCNA: Introduction to Networks, Cisco Networking Academy
- AI Prompt Design, Straits Interactive

Interests: Chess, language learning, photography, music, stargazing

PORTFOLIO

MAIze: AI-Powered Maize Disease Diagnosis System (Guidance Module) *Thesis Project*

Farmers need fast, reliable answers to crop disease questions, but general-purpose AI chatbots can generate incorrect or fabricated agricultural advice. This project developed a retrieval-augmented generation module that grounds its recommendations in a curated agricultural knowledge base, deployed as the guidance backend for a mobile diagnostic application designed for Filipino farmers.

- Retrieval pipeline querying a ChromaDB vector store of verified documents before generating a response
- Evaluated with RAGAS and a domain expert review of 30 cases, rated 4.67 out of 5 overall

Technologies: Python, Flask, LangChain, Google Gemini API, ChromaDB, RAGAS, Pydantic, Docker, Google Cloud Run
Link: **github.com/walt1210/maize_nlp**

YellowMAIze: Mobile-Deployed Maize Disease Diagnosis System *Thesis Project*

Farmers working in the field often lack reliable internet access, making diagnostic tools that depend on cloud inference impractical on site. This project designed and validated a mobile-efficient multi-task deep learning model that performs disease classification, severity estimation, and lesion segmentation from a single leaf photograph, running entirely on-device.

- Three outputs from a single forward pass in about 60 milliseconds on a CPU, via TensorFlow Lite
- Severity estimates compared with independent ratings from two team members (Spearman correlation of 0.875)

Technologies: PyTorch, TensorFlow Lite, ONNX, scikit-learn, Grad-CAM family, Cohen's Kappa
Link: **github.com/LunocoreWired/yellowmaize**

BayanBench: Auditing AI Knowledge of Philippine Towns *Independent Project, AI Evaluation*

Government offices and businesses are adopting AI chatbots, yet no public test measured what these models know about individual Philippine towns. This benchmark uses official government data as the answer key to measure accuracy and fairness across 400 towns and five models.

- Smallest towns were placed correctly as little as 11 percent of the time; Mindanao towns had half the odds of Luzon
- Official records in the prompt raised smallest-town accuracy to 97 percent or more and nearly ended invented names

Technologies: Python, pandas, NumPy, Matplotlib, Ollama, Google Gemini API, Google Colab
Link: **github.com/walt1210/bayanbench**

EventOps AI: Governed Multi-Agent Event Planning *Group Academic Project, AI Governance*

Student organizations often plan events from memory, and budgets or equipment are easily missed. This system turns a one-sentence request into an event plan through four LangGraph agents, while Python computes every cost and a human organizer must approve the result.

- Coded governance rules, prompt-injection and personal-data checks, and an append-only audit log run before any plan is shown
- Workflow pauses for human approval; 7 of 7 test cases passed, including over-budget and out-of-scope requests

Technologies: Python, LangGraph, Google Gemini API, Pydantic, pandas, ipywidgets
Link: **github.com/walt1210/eventops-ai**

Multi-Agent Report Writer *Individual Academic Project, Agentic AI*

A language model asked to summarize documents can cite passages that do not support its claims. This LangGraph system divides the work among a Router, a Reader, a Writer, and a Reviewer, and produces a cited PDF report from local documents.

- Reader gathers evidence through a custom MCP server and a ChromaDB retriever, with memory carried across tasks
- Reviewer checks every citation; in testing it caught three citations to passages never retrieved and forced a revision

Technologies: Python, LangGraph, Google Gemini API, MCP, ChromaDB, Sentence Transformers, ReportLab
Link: **github.com/walt1210/multiagent-report-writer**

Closed-Loop Text-to-SQL Agent *Academic Project, Agentic AI*

AI-generated SQL can be valid yet semantically wrong and fail silently. In a producer-critic reflection loop, a Producer model drafts each query and a Critic model checks it against the live schema before execution, with a three-pillar recovery cascade for queries that still fail.

Technologies: Python, Google Gemini API, Pydantic, SQLite
Link: **github.com/walt1210/text-to-sql-agent**

Additional projects include a two-agent CrewAI research pipeline (github.com/walt1210/multi-agent-research-pipeline), Reddit text classification and topic clustering (notebooks available on request), and a relational database design project. More at github.com/walt1210.